Report

Experimental study and construction of a pipeline for Aspect Term Extraction, Opinion Term Extraction and aspect-level sentiment

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Abstract

The project was developed with the aim of separately studying the main components of an aspect-based sentiment analysis pipeline, comparing multiple strategies for **Aspect Term Extraction** (**ATE**) and **Opinion Term Extraction** (**OTE**) before integrating the best methods into a final pipeline. For ATE, **RNN**, **LSTM** and **BiLSTM** architectures were compared, using both randomly learned embeddings and pre-trained GloVe embeddings. For OTE, a method based mainly on **POS tagging** and a method based on syntactic dependencies (**Dependency Parsing**) were compared. The final pipeline was initially evaluated on a gold standard manually created from restaurant reviews belonging to the test set; subsequently, a second evaluation was carried out on another manually created gold standard of reviews from the laptop domain: in this way, the effectiveness of the pipeline on a different domain could be analyzed. Comparison with ABSA results from the literature shows that the sentiment classifier achieves competitive values on the evaluable subsets, but the pipeline as a whole remains strongly limited by ATE generalization and by error propagation across stages.

1. Introduction

The work begins with an experimental analysis of the various components of the pipeline: separate notebooks were built for ATE and OTE with the aim of comparing different alternatives, measuring the results and selecting the configurations to be reused in the final pipeline. Finally, for each stage of the pipeline, the solutions that offered the best results were selected.

ATE was treated as a sequence labeling problem with **BIO** labels. Three types of recurrent networks **- RNN**, **LSTM** and **BiLSTM -** were compared with two different types of embeddings.

OTE was addressed with two rule-based procedures built on top of **spaCy** and **SentiWordNet**: one aimed at finding candidates through **POS tagging**, the other constrained by dependency relations among the different functions of words in the sentence (**Dependency Parsing**). After selecting the two best methods, the OTE output was used as input to a **VADER**-based sentiment module.

The final evaluation was conducted in two scenarios: the first is **in-domain** and uses restaurant reviews, consistent with the data on which the ATE model was trained. The second is **cross-domain** and uses a gold standard of laptop reviews, without retraining the model, to determine which parts of the pipeline depend most strongly on the domain. This second evaluation was not used to reselect the models, but exclusively as a generalization test.

2. Dataset, data preparation and experimentation

2.1 Restaurant dataset

The restaurant dataset used in the notebooks contains 3693 aspect term annotations distributed across 2019 unique sentences. Each row provides the sentence, the aspect term, the polarity, and the start and end character offsets of the aspect. For ATE, annotations referring to the same sentence were grouped, obtaining a single sequence per sentence with multiple possible aspects.

Tokenization was performed with **TreebankWordTokenizer** and the dataset offsets were converted into BIO labels. In the notebook dedicated to ATE, the dataset was split with seed 42 into 1615 training sentences, 202 validation sentences and 202 test sentences; in terms of aspect terms, the splits contained 2975, 330 and 385 aspects respectively. This split was maintained for the comparison among architectures, so that the metrics would be directly comparable.

2.2 Manual gold standard for OTE and the pipeline

To measure OTE and the integrated pipeline, a manual gold standard of 100 restaurant annotations was used, each consisting of **sentence**, **aspectTerm**, **opinionPhrase** and **polarity**.

2.3 Method selection criterion

The choices were not made on the basis of an a priori preference for a specific architecture. For ATE, the main criterion was F1 on the test set, while also observing precision and recall. For OTE, both an example-level success measure and precision, recall and F1 on normalized opinion phrases were considered; the solution with higher F1 and precision was preferred, even though it produced fewer overall predictions. The final pipeline was therefore built by reusing the **BiLSTM with GloVe embeddings** and **dependency-based** extraction.

3. Study of Aspect Term Extraction (ATE)

3.1 Configurations analyzed

Six configurations were implemented in the ATE notebook: RNN, LSTM and BiLSTM with randomly initialized embeddings, and the same three architectures with 300-dimensional GloVe 6B embeddings. For the pre-trained models, the embeddings were trainable during training; RNN and LSTM used a hidden size of 256, while the BiLSTM produced a bidirectional representation with an overall size of 512 before the linear layer. Dropout 0.5 was used in the pre-trained versions and in the random BiLSTM.

All models were trained with **Adam**, **learning rate 0.0005**, **cross-entropy with padding index equal to 0**, and **early stopping with patience 4**. Selection of the best checkpoint was guided by F1 on the validation set. The final evaluation was computed with **seqeval** on the BIO sequences, i.e. at the extracted-entity level.

3.2 Results with random embeddings

Model	**Precision**	**Recall**	**F1**
RNN	0,7297	0,6519	0,6886
LSTM	0,6899	0,6182	0,6521
BiLSTM	0,7927	0,6753	0,7293

Table 1. ATE results with randomly learned embeddings.

Among the models with random embeddings, the BiLSTM was clearly the best configuration: F1 0.7293 versus 0.6886 for the RNN and 0.6521 for the LSTM. The advantage derives mainly from precision, equal to 0.7927, while recall remains lower at 0.6753. Already at this stage, the BiLSTM appeared to be potentially the best choice.

3.3 Results with pre-trained GloVe

Model	**Precision**	**Recall**	**F1**
RNN + GloVe	0,6861	0,7039	0,6949
LSTM + GloVe	0,7150	0,7299	0,7224
BiLSTM + GloVe	0,7595	0,7299	0,7444

Table 2. ATE results with 300d GloVe embeddings.

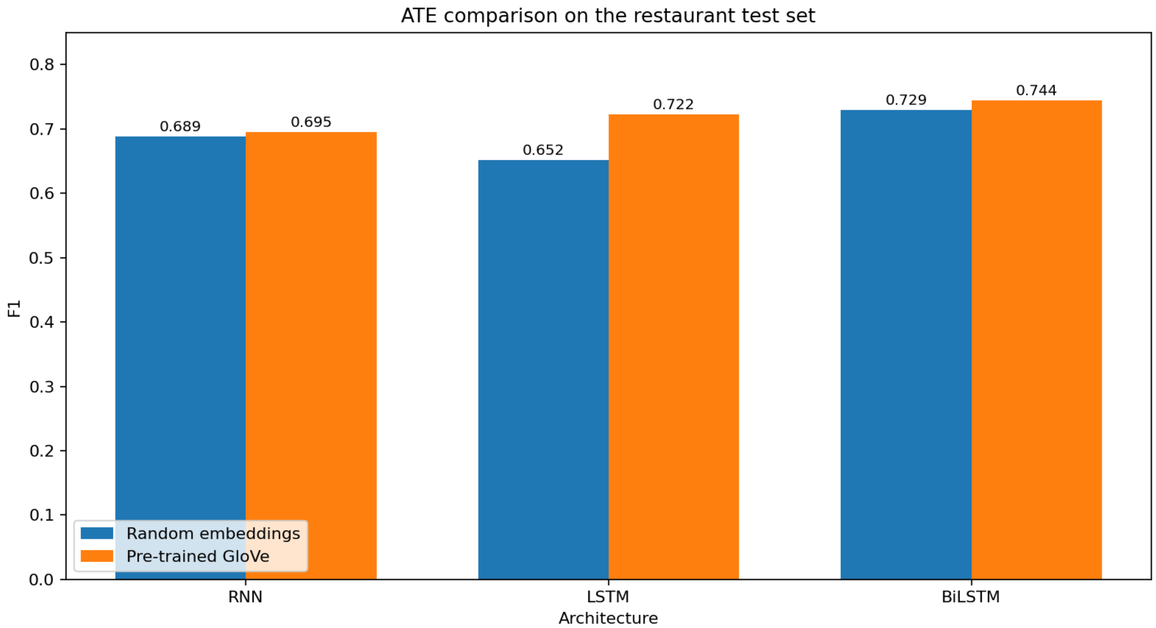

Figure 1. F1 comparison between random and pre-trained embeddings.

Introducing GloVe produces a significant improvement especially for LSTM and BiLSTM. The LSTM increases from F1 0.6521 to 0.7224, while the BiLSTM reaches the highest value of the entire study, **0.7444**. The RNN improves only marginally, from 0.6886 to 0.6949. BiLSTM + GloVe also maintains a better balance between precision (0.7595) and recall (0.7299) than the corresponding version with random embeddings.

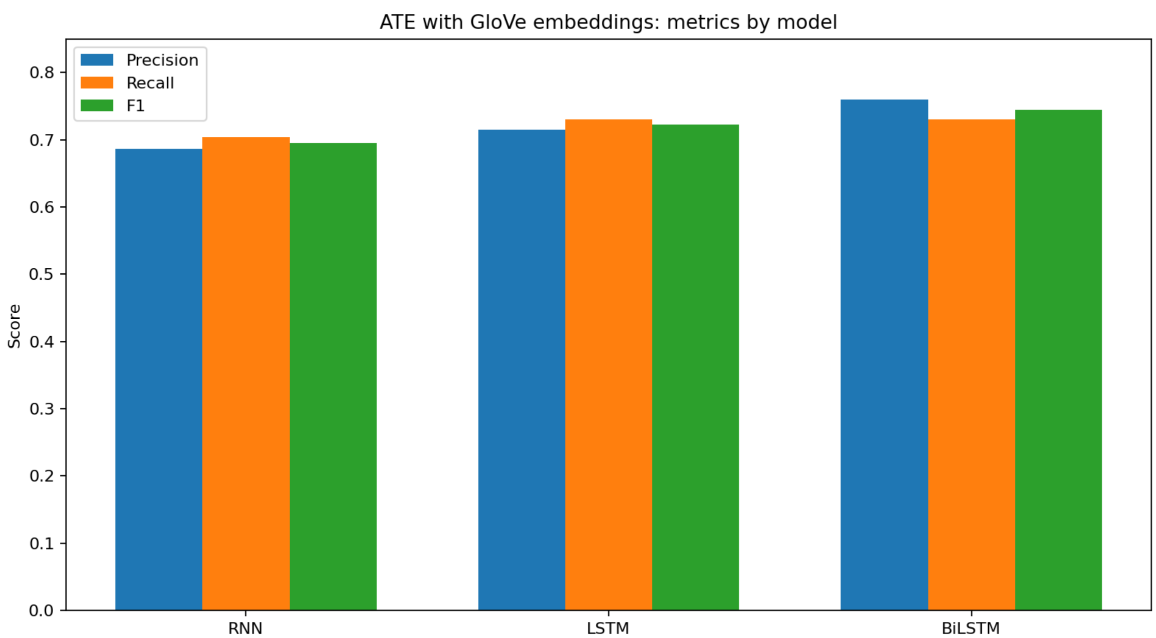

Figure 2. Precision, recall and F1 of the ATE configurations with GloVe.

3.4 Final choice for ATE

The **BiLSTM with GloVe** was selected for the pipeline because it achieves the best F1 among all configurations and does not depend on an advantage obtained on only a single metric.

4. Study of Opinion Term Extraction (OTE)

4.1 POS tagging method

The first OTE method analyzes each sentence with spaCy, identifies the position of the aspect term and searches for opinion candidates among adjectives, verbs and nouns. Candidates are filtered using a specific stop-word list and **SentiWordNet**; when SentiWordNet does not contain a lemma, adjectives are still retained as possible opinion words. Candidates are sorted by distance from the center of the aspect and at most the two nearest opinion phrases are returned. Phrase expansion incorporates relevant modifiers, negations and complements.

Applied to the 3693 restaurant annotations, the POS tagging method produces at least one opinion phrase in 3613 cases, equal to 97.83% of the dataset. However, this high coverage results in a substantial number of false positives: on the gold standard of 100 examples it obtains 89 true positives, 100 false positives and 16 false negatives, with precision 0.4709, recall 0.8476 and F1 0.6054. At the example level, 68 annotations out of 100 were considered correct.

4.2 Dependency Parsing method

The second method restricts the search to tokens syntactically connected to the aspect term: children of the aspect, the head of the aspect, and children of the head. Among these candidates, **ADJ**, **VERB** and **ADV** tokens that pass the same lexical filter as the POS-based method are considered. The opinion phrases are then expanded using the same rules and discarded when they exceed four tokens. Finally, candidates are sorted by distance.

The dependency-based procedure produces at least one opinion phrase in 2373 cases out of 3693, i.e. 64.26%. Coverage is therefore much lower than the 97.83% of the POS method, but the number of false positives decreases substantially. On the gold standard, 71 true positives, 45 false positives and 34 false negatives are observed, with precision 0.6121, recall 0.6762 and F1 0.6425. The example-level accuracy measure is 56%, lower than the 68% of POS tagging, but the opinion-term-level evaluation shows higher overall quality.

Method	**Coverage on 3693**	**Precision**	**Recall**	**F1**	**Example success**
POS-based	97,83%	0,4709	0,8476	0,6054	68%
Dependency-based	64,26%	0,6121	0,6762	0,6425	56%

Table 3. Comparison of the two OTE strategies.

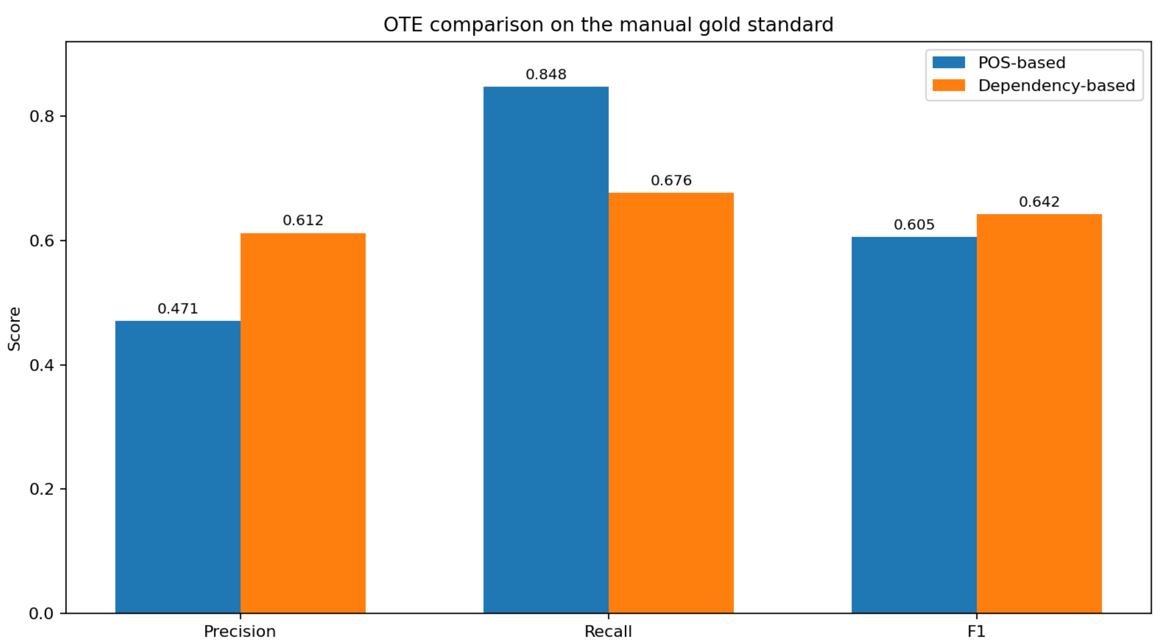

Figure 3. Precision, recall and F1 for POS-based and Dependency-based OTE.

4.3 Decision on the OTE method

Dependency Parsing was selected for the final pipeline: the decision is motivated by the fact that the pipeline must not only produce an opinion phrase, but also associate it correctly with the aspect. The POS tagging method maximizes coverage and recall, but generates many unrelated candidates and has precision below 0.5. The Dependency-based method instead reduces false positives, raises precision to 0.6121 and achieves the highest F1, 0.6425. Since OTE errors propagate directly to sentiment, reducing coverage in favor of more selective and correct associations was considered preferable.

5. Construction and evaluation of the final pipeline

5.1 Integration of the selected methods

The final pipeline reuses the BiLSTM with GloVe embeddings to extract aspect terms from sentences. For each predicted aspect, the OTE Dependency Parsing function is executed. The resulting opinion phrases are then passed to VADER; for each phrase, the compound score and the corresponding **positive**, **negative** or **neutral** class are stored using the standard ±0.05 thresholds. When an aspect has multiple opinion phrases, the final polarity is obtained by averaging the compound scores. If no opinion phrase is extracted, the sentiment for that aspect remains unavailable.

The ATE/OTE evaluation uses exact matching after normalization of sentence, aspect term and opinion phrase. ATE success indicates that the gold aspect is present among those predicted; opinion coverage measures whether at least one gold opinion was recovered when ATE is correct; end-to-end coverage instead requires complete recovery of the gold opinions. A stricter sentence-aspect-opinion-level measure was also calculated, with precision, recall and F1 on sets of pairs.

5.2 Results on the restaurant domain

Stage	**Metric**	**Score**
ATE	Success Rate	0,8333 (83,33%)
ATE → OTE	Opinion Coverage | ATE correct	0,7333 (73,33%)
ATE → OTE	End-to-End Coverage	0,6111 (61,11%)
Aspect-Opinion Pair	Precision	0,4400
Aspect-Opinion Pair	Recall	0,6111
Aspect-Opinion Pair	F1	0,5116
Sentiment	Coverage | ATE correct	0,8667 (86,67%)
Sentiment	Accuracy	0,8462 (84,62%)
Sentiment	Macro Precision	0,8000
Sentiment	Macro Recall	0,7778
Sentiment	Macro F1	0,7047

Table 4. Results of the final pipeline on the restaurant domain.

In the final notebook, the gold standard was intersected with the ATE test split: only 18 annotations actually belonged to sentences in the test set, therefore all final restaurant metrics involving ATE, OTE and sentiment refer to these instances. Of the 18 gold annotations in the test split, 15 aspects are correctly recognized, corresponding to an ATE success rate of 83.33%. Among these cases, OTE recovers at least one gold opinion in 73.33% of the instances. Considering the entire chain and requiring all gold opinions to be covered, success falls to 61.11%. The pair-based evaluation also shows 11 true positives, 14 false positives and 7 false negatives, with F1 0.5116.

The ATE/OTE error analysis divides the 18 instances into 11 complete successes (61.11%), 4 OTE failures (22.22%) and 3 ATE failures (16.67%). The result shows that, in the training domain, the main degradation stage is not ATE but opinion-term association. This is consistent with the greater difficulty of OTE also observed in the separate notebook.

5.3 Sentiment evaluation and error analysis

Of the 15 instances with a correct aspect term, 13 receive a sentiment prediction, for coverage of 86.67%. On these 13 instances, accuracy is 84.62% and Macro-F1 is 0.7047; the classification report shows F1 0.95 for positive, 0.50 for negative and 0.67 for neutral. The positive class is clearly predominant in the small evaluable subset, with 9 examples out of 13.

The error analysis highlights different limitations. For the aspect 'Halibut', OTE correctly extracts 'too salty', but VADER assigns a compound score of 0.0 and therefore neutral, while the gold label is negative: the error is attributable to the sentiment analyzer's lexicon. For 'dessert', instead, the predicted opinion phrase is "n't waste poor", which combines material referring to different segments of the sentence and leads VADER to a positive polarity; in this case the main source is the OTE association.

6. Cross-domain evaluation on laptop reviews

6.1 Setup

To verify the robustness of the pipeline, the same ATE model trained on restaurants was applied to a laptop gold standard consisting of 276 annotations. The laptop dataset was not used for training, tuning or selection of the OTE rules. The evaluation therefore represents a scenario with a completely different domain.

For each annotation, **sentence**, **aspectTerm**, **opinionPhrase** and **polarity** were available; matching and metrics were computed using the same functions used on the gold standard for the restaurant dataset.

6.2 Results

Stage	Metric	Restaurant	Laptop
ATE	Success Rate	0,8333	0,3261
ATE → OTE	Opinion Coverage | ATE correct	0,7333	0,6556
ATE → OTE	End-to-End Coverage	0,6111	0,2138
Aspect-Opinion Pair	F1	0,5116	0,2830
Sentiment	Coverage | ATE correct	0,8667	0,8333
Sentiment	Accuracy	0,8462	0,8800
Sentiment	Macro F1	0,7047	0,8136

Table 5. Comparison between restaurant evaluation and cross-domain laptop evaluation.

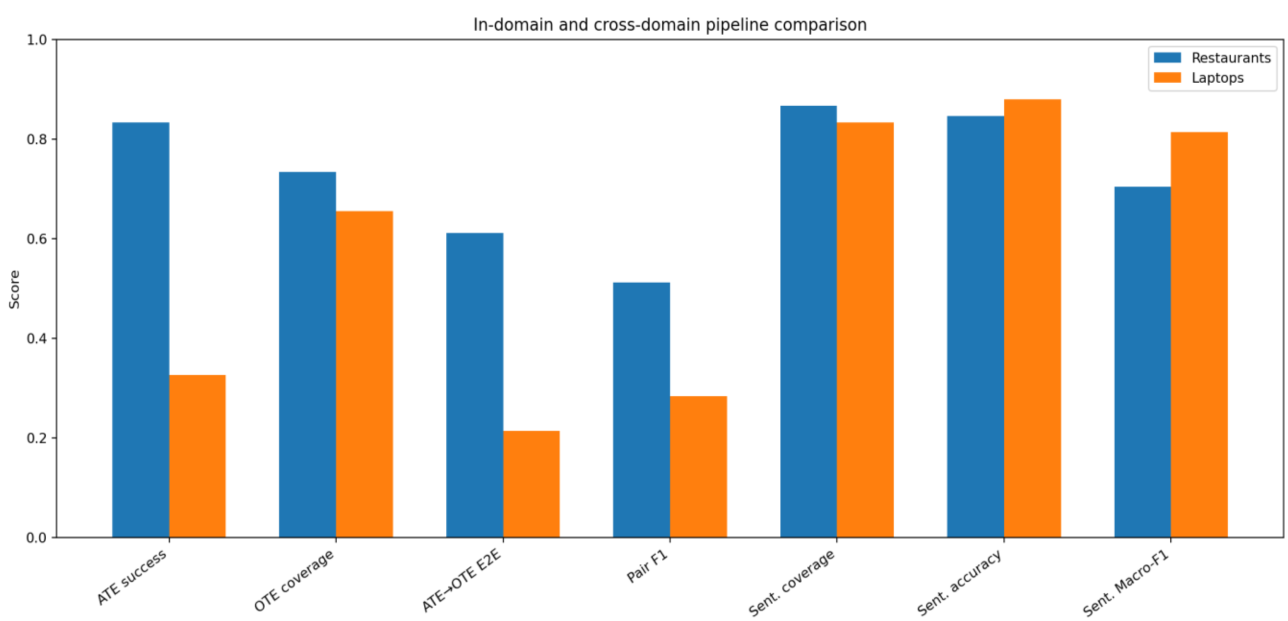

Figure 4. Comparison of pipeline metrics in the two domains.

The most evident result is the collapse of ATE: the success rate drops from 83.33% to 32.61%. Out of 276 laptop annotations, 186 are classified as ATE failures, 59 as complete ATE→OTE successes and 31 as OTE failures. End-to-end coverage consequently falls to 21.38%. The F1 of aspect-opinion pairs also decreases from 0.5116 to 0.2830; on laptops, 59 true positives, 82 false positives and 217 false negatives were measured.

OTE instead shows a more limited degradation when conditioned on correct ATE: opinion coverage decreases from 73.33% to 65.56%. This behavior suggests that the dependency-based strategy exploits linguistic relations that transfer more effectively across domains than the vocabulary and patterns learned by ATE: this means that the cross-domain problem is mainly related to aspect extraction.

Sentiment maintains very similar coverage (86.67% restaurant versus 83.33% laptop) and, on valid cases, achieves 88.00% accuracy and Macro-F1 0.8136. The result is higher than for restaurants, but 75 sentiment predictions are evaluated out of 90 correctly recognized aspect terms, while all ATE failures are excluded; therefore, the metric describes only the quality of sentiment prediction when the pipeline has successfully reached the sentiment analysis stage.

6.3 Cross-domain interpretation

The laptop experiment is particularly useful because it separates the effect of the domain on the modules. The BiLSTM was trained only on restaurant reviews and therefore tends to recognize lexical and contextual patterns typical of that domain more reliably. When moving to laptops, the number of unrecognized aspect terms increases considerably. OTE and VADER, on the other hand, are procedures less dependent on restaurant training: for this reason, their conditional metrics decrease less or, in the case of sentiment, improve on the evaluated subset.

7. Comparison with ABSA methods from the literature

7.1 Comparison protocol

The comparison with the literature for the ABSA method is limited to polarity classification for an already known aspect term. The reference results are taken from works using the SemEval-2014 Laptop and Restaurant datasets: ATAE-LSTM by Wang et al. (2016), BERT-SPC and LCF-BERT in the evaluation by Zeng et al. (2019) are reported.

Method	Laptop Accuracy	Restaurant Accuracy	Note
ATAE-LSTM	68,70%	77,20%	Aspect-level sentiment classification
BERT-SPC	80,25%	85,98%	BERT with aspect/sentence pair
LCF-BERT (CDW)	82,45%	87,14%	Local Context Focus, best reported variant
Proposed pipeline*	88,00%	84,62%	Only cases with correct ATE and available sentiment

Table 6. Indicative comparison with ABSA results on SemEval-2014

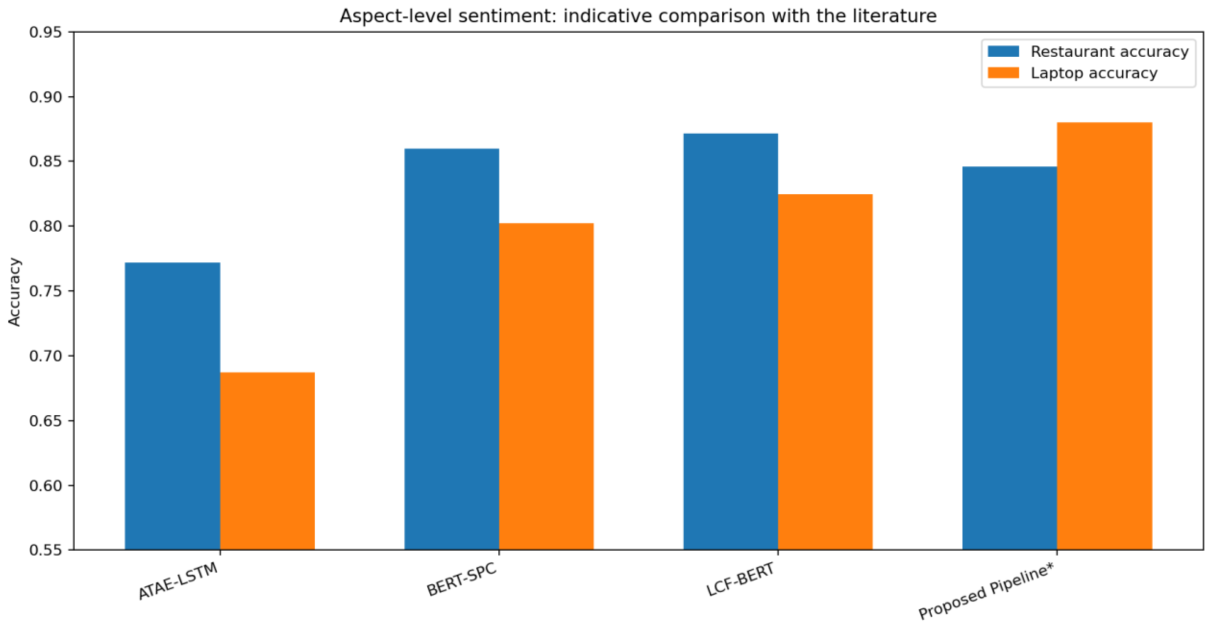

Figure 5. Sentiment accuracy: indicative comparison with published ABSA models. *Metrics are not directly comparable.

7.2 Discussion of the comparison

On the evaluable restaurant subset, the pipeline accuracy (84.62%) is higher than the value reported for ATAE-LSTM (77.20%), close to BERT-SPC (85.98%) and lower than LCF-BERT CDW (87.14%). On the laptop gold standard, the pipeline reaches 88.00% on the 75 cases with a correct aspect and available sentiment, a value higher than the three cases considered; however, it is important to take into account that the proposed pipeline is more selective and discards both ATE failures and cases in which OTE does not produce sufficient material for VADER.

The most useful comparison is therefore qualitative. ABSA models from the literature achieve high accuracy when they directly receive the target aspect and learn sentiment using the sentence context. The proposed pipeline shows that a lightweight lexical method can achieve good results once the sentiment stage is correctly reached, but its overall effectiveness depends on the ability to extract the aspect and associate an appropriate opinion phrase.

8. Conclusions and limitations

The project achieved its main objective of separately analyzing ATE and OTE, selecting the best methods based on the results and verifying their behavior when chained together. For ATE, the comparison among six configurations showed that the BiLSTM with GloVe embeddings is the most effective solution, with F1 0.7444 in the comparative notebook. For OTE, the dependency-based method was preferred to POS tagging despite its lower coverage, because it offers higher precision and F1 and reduces false positives.

The integration of these two modules with VADER produced, on the restaurant subset, 83.33% ATE success, 73.33% OTE coverage, 61.11% ATE→OTE success and 84.62% sentiment accuracy on valid cases.

The test on the laptop dataset showed that the pipeline is not fully capable of operating out of domain: ATE drops to 32.61%, while OTE given correct ATE remains at 65.56% and sentiment reaches 88.00% accuracy. The behavior confirms that the component most sensitive to the domain is the sequence-labeling model trained on restaurant reviews; the dependency-based procedure and lexical sentiment are more transferable, while still depending on the correctness of the preceding stages.

8.1 Limitations

The project has some limitations that must be considered when interpreting the results obtained. First, the gold standard used for evaluation on the restaurant domain is rather small. Of the 100 manual annotations available overall, only 18 belong to sentences present in the test split and can therefore be used for a consistent evaluation.

The method selected for OTE also introduces some limitations. The approach is rule-based and depends on the syntactic analysis performed by spaCy and on the information provided by SentiWordNet. Errors in dependency parsing can therefore propagate to the extraction and association of opinion terms. The problem is particularly evident in the presence of more complex syntactic structures, coordinations or discontinuous opinion phrases.

Similar limitations are present in the sentiment analysis stage. VADER is applied to the previously extracted opinion phrases and, consequently, does not necessarily have access to the full context of the original sentence. Some expressions may therefore receive a polarity different from the one expected in the specific domain.

Furthermore, the cross-domain evaluation on laptops highlights an important limitation related to ATE generalization ability. The model used in the final pipeline was trained exclusively on the restaurant domain and no multi-domain training procedure was carried out.

Finally, an additional difficulty concerns the very nature of OTE: for each aspect term present in a sentence, it is not always possible to identify an explicit and therefore delimited opinion phrase. In some cases, the judgment toward an aspect emerges from the overall meaning of the sentence, from a more elaborate expression, or from contextual information, without being attributable to a specific evaluative term. This characteristic introduces a certain ambiguity both in the construction of the gold standard and in the automatic evaluation of OTE.

8.2 Possible developments

The most natural developments, while maintaining the same experimental setup, are: expanding the restaurant gold standard and balancing polarities. In addition, it could be useful to introduce an ATE model trained on multiple domains and improve the OTE stage, including explicit handling of coordinations and negations.

References

[1] *Y. Wang*, *M. Huang*, *X. Zhu*, *L. Zhao*, *'Attention-based LSTM for Aspect-level Sentiment Classification'*, *2016*.

[2] *B. Zeng*, *H. Yang*, *R. Xu*, *W. Zhou*, *X. Han*, *'LCF: A Local Context Focus Mechanism for Aspect-Based Sentiment Classification'*, *2019*.